

Ch37: 3, 10, 11, 22, 23, 30, 43, 59

Basic score 4` + problem 11 2'+problem 23 2'+ problem 59 2'

3. (a) In frame S , our coordinates are such that $x_1 = +1200$ m for the big flash, and $x_2 = 700$ m for the small flash (which occurred later). Thus,

$$\Delta x = x_2 - x_1 = -500 \text{ m.}$$

If we set $\Delta x' = 0$ in Eq. 37-25, we find

$$0 = \gamma(\Delta x - v\Delta t) = \gamma(-500\text{m} - v(5.00 \times 10^{-6}\text{s}))$$

which yields $v = -1.00 \times 10^8$ m/s, $\beta = v/c = 0.33$

(b) The negative sign in part (a) implies that frame S' must be moving in the $-x$ direction.

(c) (d) Equation 37-28 leads to

$$\Delta t' = \gamma \left(\Delta t - \frac{v\Delta x}{c^2} \right) = \frac{5.00 \times 10^{-6}\text{s} - \frac{(-1.00 \times 10^8\text{m/s})(-500\text{m})}{(3 \times 10^8\text{m/s})^2}}{\sqrt{1 - \left(\frac{1 \times 10^8\text{m/s}}{3 \times 10^8\text{m/s}} \right)^2}} = 4.7 \times 10^{-6}\text{s},$$

which turns out to be positive (regardless of the specific value of γ). Thus, the order of the flashes is the same in the S' frame as it is in the S frame (where Δt is also positive). Thus, the big flash occurs first, and the small flash occurs later.

10. (a) Equation 37-36 leads to

$$\begin{aligned} \lambda &= \sqrt{\frac{c+v}{c-v}} \lambda_0 \\ \lambda &= 525\text{nm}, \quad \lambda_0 = 513\text{nm} \\ v &= 6.94 \times 10^6 \text{ m/s} \end{aligned}$$

(a) The line is shifted to a larger wavelength, which means shorter frequency. Recalling Eq. 37-31 and the discussion that follows it, this means galaxy NGC is moving away from Earth.

(b) Red shift

11. Let S be the reference frame of the micrometeorite, and S' be the reference frame of the spaceship. We assume S to be moving in the $+x$ direction. Let u be the velocity of the

micrometeorite as measured in S and v be the velocity of S' relative to S , the velocity of the micrometeorite as measured in S' can be solved by using:

$$u = \frac{u' + v}{1 + u'v/c^2} \Rightarrow u' = \frac{u - v}{1 - uv/c^2}.$$

The problem indicates that $v = -0.94c$ (spaceship velocity) and $u = +0.94c$ (micrometeorite velocity). We solve for the velocity of the micrometeorite relative to the spaceship:

$$\begin{aligned} u' &= \frac{u - v}{1 - uv/c^2} && \text{0.5 point} \\ &= \frac{0.94c - (-0.94c)}{1 - (0.94)(-0.94)} = 0.998c && \text{正确带入 u and v 0.5 point} \end{aligned}$$

we conclude that observers on the ship measure a transit time for the micrometeorite (as it passes along the length of the ship) equal to

$$\begin{aligned} \Delta t &= \frac{d}{u'} && \text{0.5 point} \\ &= \frac{280\text{m}}{0.998 \times 3.0 \times 10^8 \text{ m/s}} = 0.94 \times 10^{-6} \text{ s} && \text{0.5 point} \end{aligned}$$

22. The “coincidence” of $x = x' = 0$ at $t = t' = 0$ is important for Eq. 37-21 to apply without additional terms. In part (a), we apply these equations directly with

$$v = +0.400c = 1.199 \times 10^8 \text{ m/s},$$

and in part (c) we simply change $v \rightarrow -v$ and recalculate the primed values.

(a) The position coordinate measured in the S' frame is

$$x' = \gamma(x - vt) = \frac{x - vt}{\sqrt{1 - \beta^2}} = \frac{3.00 \times 10^8 \text{ m} - (1.199 \times 10^8 \text{ m/s})(1.50 \text{ s})}{\sqrt{1 - (0.400)^2}} = 1.3 \times 10^8 \text{ m},$$

(b) The time coordinate measured in the S' frame is

$$t' = \gamma\left(t - \frac{vx}{c^2}\right) = \frac{t - \beta x/c}{\sqrt{1 - \beta^2}} = \frac{1.50 \text{ s} - (0.400)(3.00 \times 10^8 \text{ m})/2.998 \times 10^8 \text{ m/s}}{\sqrt{1 - (0.400)^2}} = 1.20 \text{ s}.$$

(c) Now, we obtain

$$x' = \frac{x + vt}{\sqrt{1 - \beta^2}} = \frac{3.00 \times 10^8 \text{ m} + (1.199 \times 10^8 \text{ m/s})(1.50 \text{ s})}{\sqrt{1 - (0.400)^2}} = 5.24 \times 10^8 \text{ m}.$$

(d) Similarly,

$$t' = \gamma \left(t + \frac{vx}{c^2} \right) = \frac{1.50 \text{ s} + (0.400)(3.00 \times 10^8 \text{ m}) / 2.998 \times 10^8 \text{ m/s}}{\sqrt{1 - (0.400)^2}} = 2.07 \text{ s}.$$

23. (a) We take the flashbulbs to be at rest in frame S , and let frame S' be the rest frame of the second observer. Clocks in neither frame measure the proper time interval between the flashes, so the full Lorentz transformation (Eq. 37-21) must be used. Let t_s be the time and x_s be the coordinate of the small flash, as measured in frame S . Then, the time of the small flash, as measured in frame S' , is

$$t'_s = \gamma \left(t_s - \frac{\beta x_s}{c} \right)$$

where $\beta = v/c = 0.450$ and

$$\gamma = 1 / \sqrt{1 - \beta^2} = 1 / \sqrt{1 - (0.450)^2} = 1.1198 \quad \text{0.5 point}$$

Similarly, let t_b be the time and x_b be the coordinate of the big flash, as measured in frame S . Then, the time of the big flash, as measured in frame S' , is

$$t'_b = \gamma \left(t_b - \frac{\beta x_b}{c} \right).$$

Subtracting the second Lorentz transformation equation from the first and recognizing that $t_s = t_b$ (since the flashes are simultaneous in S), we find

$$\begin{aligned} \Delta t' = t'_b - t'_s &= \frac{\gamma \beta (x_s - x_b)}{c} && \text{0.5 point} \\ &= \frac{(1.1198)(0.450)(30 \times 10^3 \text{ m})}{3.00 \times 10^8 \text{ m/s}} = 5.04 \times 10^{-5} \text{ s} && \text{0.5 point} \end{aligned}$$

(b) Since $\Delta t'$ is positive, t'_b is greater than t'_s . The small flash occurs first in S' . 0.5 point

30. (a) We use Eq. 37-29:

$$u = \frac{u' + v}{1 + u'v/c^2} = \frac{0.420c + 0.620c}{1 + (0.420)(0.620)} = 0.825c,$$

in the direction of increasing x (since $v > 0$). In unit-vector notation, we have $\vec{v} = (0.825c)\hat{i}$.

(b) The classical theory predicts that $v = 0.420c + 0.620c = 1.040c$, or $\vec{v} = (1.040c)\hat{i}$.

(c) Now $v' = -0.42c\hat{i}$ so

$$u = \frac{u' + v}{1 + u'v/c^2} = \frac{-0.420c + 0.620c}{1 + (-0.420)(0.620)} = 0.270c ,$$

or $\vec{v} = (0.27c)\hat{i}$

(d) By contrast, the classical prediction is $v = 0.620c - 0.420c = 0.200c$, or $\vec{v} = (0.200c)\hat{i}$.

43. (a) The speed of the traveler is $v = 0.9950c$, which may be equivalently expressed as 0.9950 ly/y . Let d be the distance traveled. Then, the time for the trip as measured in the frame of Earth is

$$\Delta t = d/v = (26.00 \text{ ly})/(0.9950 \text{ ly/y}) = 26.13 \text{ y}.$$

(b) The signal, presumed to be a radio wave, travels with speed c and so takes 26.0 y to reach Earth. The total time elapsed, in the frame of Earth, is

$$26.13 \text{ y} + 26.0 \text{ y} = 52.13 \text{ y} .$$

(c) The proper time interval is measured by a clock in the spaceship, so $\Delta t_0 = \Delta t/\gamma$. Now

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}} = \frac{1}{\sqrt{1 - (0.9950)^2}} = 10.01$$

Thus, $\Delta t_0 = (26.13 \text{ y})/(10.01) = 2.610 \text{ y}$.

59. In the laboratory, it travels a distance

$$d = 0.856 \text{ mm} = vt \quad \text{0.5 point}$$

where $v = 0.992c$ and t is the time measured on the laboratory clocks. We can relate t to the proper lifetime of the particle t_0 :

$$\begin{aligned} t &= \frac{t_0}{\sqrt{1 - (v/c)^2}} \Rightarrow t_0 = t \sqrt{1 - \left(\frac{v}{c}\right)^2} & \text{0.5 point} \\ &= \frac{d}{0.992c} \sqrt{1 - 0.992^2} & \text{0.3 point} \end{aligned}$$

which yields $t_0 = 3.63 \times 10^{-13} \text{ s} = 0.363 \text{ ps}$. **0.2 point**